In [ ]:
```
# EXPERIMENT - 7: HANDLING OUTLIERS & MISSING VALUES
PIN No.: 241UDS0111

## Aim
To preprocess a student-performance dataset by removing empty rows and duplicates, detecting outliers using the
IQR method, and handling missing values with Pandas and Scikit-learn `SimpleImputer`.
```

In [ ]:
```
## Step 1: Import the basic libraries
```

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer

# Use a consistent plot style
sns.set_theme(style="whitegrid")
```

In [ ]:
```
## Step 2: Load and analyze the dataset
```

In [2]:
```
# Load the student dataset
df_241UDS0111 = pd.read_csv("Experiment_7_241UDS0111_Student_Performance_Missing_Outliers.csv")

# Dataset information
df_241UDS0111.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 123 entries, 0 to 122
Data columns (total 24 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Branch                  121 non-null    str
 1   Marks[10th]             121 non-null    float64
 2   Marks[12th]             121 non-null    float64
 3   Gender                  121 non-null    str
 4   Board[10th]             120 non-null    str
 5   Board[12th]             121 non-null    str
 6   Category                121 non-null    str
 7   GPA 1                   121 non-null    float64
 8   Rank                    122 non-null    float64
 9   Normalized Rank         121 non-null    float64
 10  CGPA                    121 non-null    float64
 11  Current Back            121 non-null    float64
 12  Ever Back               120 non-null    float64
 13  GPA 2                   121 non-null    float64
 14  GPA 3                   121 non-null    float64
 15  GPA 4                   121 non-null    float64
 16  GPA 5                   121 non-null    float64
 17  GPA 6                   121 non-null    float64
 18  Olympiads Qualified     121 non-null    float64
 19  Technical Projects      121 non-null    float64
 20  Tech Quiz               121 non-null    float64
 21  Engg. Coaching          121 non-null    float64
 22  NTSE Scholarships       121 non-null    float64
 23  Miscellany Tech Events  121 non-null    float64
dtypes: float64(19), str(5)
memory usage: 23.2 KB
```

In [3]:
```
# Display the first five records
df_241UDS0111.head()
```

Out[3]: Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	NaN	84.50	78.77	Male	BSEB Patna	CBSE	ST
1	ECE	79.75	75.45	Female	CBSE	BSEB Patna	SC
2	IT	68.33	59.62	Male	BSEB Patna	CBSE	SC
3	CIVIL	75.58	80.85	Female	ICSE	SSC	OBC
4	CSE	52.00	62.97	Male	CBSE	SSC	GEN

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	7.54	38811.0	78.9259	...	7.99	8.22	7.94
1	6.39	39350.0	80.0220	...	6.57	6.80	5.98
2	7.82	29935.0	60.8757	...	8.13	7.65	7.90
3	6.63	36601.0	74.4316	...	6.89	6.26	6.35
4	6.18	26354.0	53.5934	...	6.03	6.55	6.03

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	7.29	1.0	2.0	1.0	1.0	1.0	0.0
1	6.42	0.0	5.0	0.0	0.0	1.0	1.0
2	8.16	1.0	2.0	1.0	1.0	0.0	1.0
3	6.23	1.0	1.0	3.0	1.0	1.0	2.0
4	6.00	0.0	2.0	1.0	0.0	1.0	1.0

5 rows × 24 columns

In [4]:
```
# Check the original shape
df_241UDS0111.shape
```

Out[4]: (123, 24)

In [5]:
```
# Remove completely empty rows
df_241UDS0111 = df_241UDS0111.dropna(how="all")
df_241UDS0111.shape
```

Out[5]: (122, 24)

In [ ]:
```
## Step 3: Find and remove duplicates
```

In [6]:
```
# Count duplicate rows
df_241UDS0111.duplicated().sum()
```

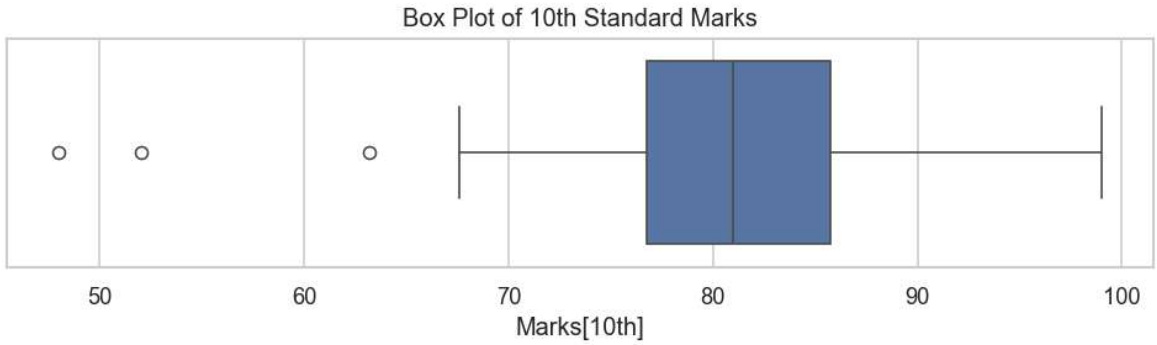
Out[6]: np.int64(2)

In [7]:
```
# Remove duplicate rows
df_241UDS0111 = df_241UDS0111.drop_duplicates().copy()
df_241UDS0111.shape
```

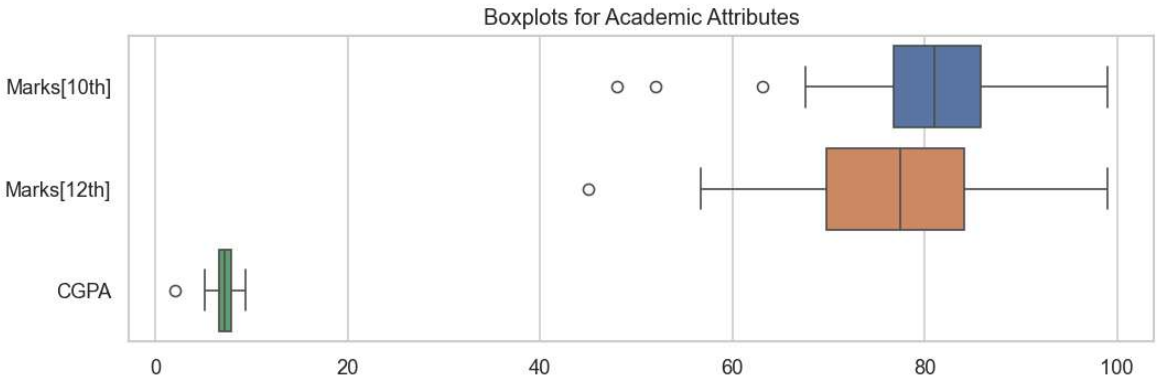
Out[7]: (120, 24)

In [8]:
```
## Step 4: Identify and remove outliers
```

In [9]:
```
# Visualize outliers in 10th-standard marks
plt.figure(figsize=(10, 2), dpi=120)
sns.boxplot(data=df_241UDS0111, x="Marks[10th]")
plt.title("Box Plot of 10th Standard Marks")
plt.show()
```



In [10]:
```
# Compare distributions of the three main academic columns
outlier_columns_241UDS0111 = ["Marks[10th]", "Marks[12th]", "CGPA"]
plt.figure(figsize=(10, 3), dpi=120)
sns.boxplot(data=df_241UDS0111[outlier_columns_241UDS0111], orient="h")
plt.title("Boxplots for Academic Attributes")
plt.show()
```



In [11]:
```
# Calculate Q1, Q3, and IQR
Q1_241UDS0111 = df_241UDS0111[outlier_columns_241UDS0111].quantile(0.25)
Q3_241UDS0111 = df_241UDS0111[outlier_columns_241UDS0111].quantile(0.75)
IQR_241UDS0111 = Q3_241UDS0111 - Q1_241UDS0111

print("Q1:\n", Q1_241UDS0111)
print("\nQ3:\n", Q3_241UDS0111)
print("\nIQR:\n", IQR_241UDS0111)
```

```
Q1:
 Marks[10th]    76.785
Marks[12th]    69.735
CGPA            6.595
Name: 0.25, dtype: float64

Q3:
 Marks[10th]    85.775
Marks[12th]    84.130
CGPA            7.935
Name: 0.75, dtype: float64

IQR:
 Marks[10th]     8.990
Marks[12th]    14.395
CGPA            1.340
dtype: float64
```

In [12]:
```
# Flag rows with an outlier in any selected column
outliers_241UDS0111 = (
    (df_241UDS0111[outlier_columns_241UDS0111] < (Q1_241UDS0111 - 1.5 * IQR_241UDS0111))
    | (df_241UDS0111[outlier_columns_241UDS0111] > (Q3_241UDS0111 + 1.5 * IQR_241UDS0111))
)
print("Outlier counts by column:\n", outliers_241UDS0111.sum())
print("Rows containing at least one outlier:", outliers_241UDS0111.any(axis=1).sum())
```

```
Outlier counts by column:
 Marks[10th]    3
Marks[12th]    1
CGPA           1
dtype: int64
Rows containing at least one outlier: 5
```

In [13]:
```
# Remove rows with outliers in the selected academic columns
df_241UDS0111 = df_241UDS0111.loc[~outliers_241UDS0111.any(axis=1)].copy()
print("Shape after outlier removal:", df_241UDS0111.shape)
```

```
Shape after outlier removal: (115, 24)
```

In [ ]:
```
## Step 5: Fill the missing values
```

In [14]:
```
# Count missing values in each column
df_241UDS0111.isnull().sum()
```

Out[14]:
```
Branch                   1
Marks[10th]              1
Marks[12th]              1
Gender                   1
Board[10th]              1
Board[12th]              1
Category                 1
GPA 1                    1
Rank                     0
Normalized Rank          1
CGPA                     1
Current Back             0
Ever Back                2
GPA 2                    1
GPA 3                    1
GPA 4                    1
GPA 5                    1
GPA 6                    1
Olympiads Qualified      1
Technical Projects       1
Tech Quiz                1
Engg. Coaching           1
NTSE Scholarships        1
Miscellany Tech Events   1
dtype: int64
```

In [15]:
```
# Fill missing values in 10th-standard marks with the mean
df_241UDS0111["Marks[10th]"] = df_241UDS0111["Marks[10th]"].fillna(df_241UDS0111["Marks[10th]"].mean())
print("Missing values in Marks[10th]:", df_241UDS0111["Marks[10th]"].isnull().sum())
```

```
Missing values in Marks[10th]: 0
```

In [16]:
```
# Use median imputation for selected numeric columns
numeric_columns_241UDS0111 = df_241UDS0111.select_dtypes(include=["number"]).columns.tolist()
numerical_imputer_241UDS0111 = SimpleImputer(strategy="median")
df_241UDS0111[numeric_columns_241UDS0111] =
numerical_imputer_241UDS0111.fit_transform(df_241UDS0111[numeric_columns_241UDS0111])

# Confirm missing numeric values are handled
df_241UDS0111[numeric_columns_241UDS0111].isnull().sum().sum()
```

Out[16]:
```
np.int64(0)
```

In [17]:
```
# Use the most frequent value to fill missing categorical columns
categorical_columns_241UDS0111 = df_241UDS0111.select_dtypes(exclude=["number"]).columns.tolist()
categorical_imputer_241UDS0111 = SimpleImputer(strategy="most_frequent")
df_241UDS0111[categorical_columns_241UDS0111] =
categorical_imputer_241UDS0111.fit_transform(df_241UDS0111[categorical_columns_241UDS0111])

# Check missing values in categorical columns
df_241UDS0111[categorical_columns_241UDS0111].isnull().sum()
```

Out[17]:
```
Branch         0
Gender         0
Board[10th]    0
Board[12th]    0
Category       0
dtype: int64
```

In [ ]:
```
## Final check: Cleaned dataset
```

In [18]:
```
# Dataset information after cleaning
df_241UDS0111.info()
```

```
<class 'pandas.DataFrame'>
Index: 115 entries, 0 to 119
Data columns (total 24 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   Branch                 115 non-null    str
 1   Marks[10th]            115 non-null    float64
 2   Marks[12th]            115 non-null    float64
 3   Gender                 115 non-null    str
 4   Board[10th]            115 non-null    str
 5   Board[12th]            115 non-null    str
 6   Category               115 non-null    str
 7   GPA 1                  115 non-null    float64
 8   Rank                   115 non-null    float64
 9   Normalized Rank        115 non-null    float64
 10  CGPA                   115 non-null    float64
 11  Current Back           115 non-null    float64
 12  Ever Back              115 non-null    float64
 13  GPA 2                  115 non-null    float64
 14  GPA 3                  115 non-null    float64
 15  GPA 4                  115 non-null    float64
 16  GPA 5                  115 non-null    float64
 17  GPA 6                  115 non-null    float64
 18  Olympiads Qualified    115 non-null    float64
 19  Technical Projects     115 non-null    float64
 20  Tech Quiz              115 non-null    float64
 21  Engg. Coaching         115 non-null    float64
 22  NTSE Scholarships      115 non-null    float64
 23  Miscellany Tech Events  115 non-null    float64
dtypes: float64(19), str(5)
memory usage: 22.5 KB
```

In [19]:
```
# Check all remaining missing values
df_241UDS0111.isnull().sum()
```

Out[19]:
```
Branch                   0
Marks[10th]              0
Marks[12th]              0
Gender                   0
Board[10th]              0
Board[12th]              0
Category                 0
GPA 1                    0
Rank                     0
Normalized Rank          0
CGPA                     0
Current Back             0
Ever Back                0
GPA 2                    0
GPA 3                    0
GPA 4                    0
GPA 5                    0
GPA 6                    0
Olympiads Qualified      0
Technical Projects       0
Tech Quiz                0
Engg. Coaching           0
NTSE Scholarships        0
Miscellany Tech Events   0
dtype: int64
```

In [20]:
```
# Display cleaned data and save a new CSV
display(df_241UDS0111.head())
cleaned_csv_241UDS0111 = "Experiment_7_241UDS0111_Student_Performance_Cleaned.csv"
df_241UDS0111.to_csv(cleaned_csv_241UDS0111, index=False)
print("Cleaned dataset saved as:", cleaned_csv_241UDS0111)
```

Columns 1–8 of 22

	Branch	Marks[10th]	Marks[12th]	Gender	Board[10th]	Board[12th]	Category
0	CSE	84.50	78.77	Male	BSEB Patna	CBSE	ST
1	ECE	79.75	75.45	Female	CBSE	BSEB Patna	SC
2	IT	68.33	59.62	Male	BSEB Patna	CBSE	SC
3	CIVIL	75.58	80.85	Female	ICSE	SSC	OBC
5	CSE	75.78	65.36	Male	SSC	SSC	OBC

Columns 9–15 of 22 · first column repeated

	GPA 1	Rank	Normalized Rank	...	GPA 3	GPA 4	GPA 5
0	7.54	38811.0	78.9259	...	7.99	8.22	7.94
1	6.39	39350.0	80.0220	...	6.57	6.80	5.98
2	7.82	29935.0	60.8757	...	8.13	7.65	7.90
3	6.63	36601.0	74.4316	...	6.89	6.26	6.35
5	6.71	31206.0	63.4604	...	7.01	6.73	6.91

Columns 16–22 of 22 · first column repeated

	GPA 6	Olympiads Qualified	Technical Projects	Tech Quiz	Engg. Coaching	NTSE Scholarships	Miscellany Tech Events
0	7.29	1.0	2.0	1.0	1.0	1.0	0.0
1	6.42	0.0	5.0	0.0	0.0	1.0	1.0
2	8.16	1.0	2.0	1.0	1.0	0.0	1.0
3	6.23	1.0	1.0	3.0	1.0	1.0	2.0
5	6.62	0.0	1.0	3.0	0.0	0.0	1.0

5 rows × 24 columns

```
Cleaned dataset saved as: Experiment_7_241UDS0111_Student_Performance_Cleaned.csv
```

In [ ]:
```
## Result
The student-performance dataset was preprocessed by removing completely empty rows and duplicates, detecting
outliers using the IQR rule, and imputing missing numerical and categorical values with median and most-frequent
strategies. The cleaned dataset was saved as a CSV file.
```